Instrumental Variables

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Where we are

- For SOO, we leaned on conditional ignorability: find an X that closes all back-doors, then adjust for it.
- For DID, we traded ignorability in levels for ignorability in trends (parallel-trends)
- Today, a different bargain: Try to find an "encouragement" to treatment, which only matters through treatment, identifying an effect in a sub-group.

The intuition: a randomized encouragement

Suppose we can't randomize *pizza* (IRB won't allow withholding pizza), but we *can* randomize a *gift card* for pizza.

Half the sample gets a card; we observe who actually eats pizza, then measure happiness.

Good news: we have an experiment with respect to the card.

- $\mathbb{E}[\text{happy} \mid \text{card}] - \mathbb{E}[\text{happy} \mid \text{no card}]$ is identified.
  This is the intent-to-treat (ITT) or reduced form effect of the card.
- But it isn't the effect of pizza. How do we get back to that?

Backing it out: four kinds of people

Imagine four groups defined by how they respond to the encouragement:

- *Never-takers*: don't eat pizza either way. Card has no effect on their pizza intake $\Rightarrow$ no effect on their happiness.
- *Always-takers*: eat pizza either way. Same: card doesn't change anything for them.
- *Compliers*: eat pizza *iff* they get the card. For them, card means pizza, no card means no pizza: effect of card (ITT) is effect of pizza.
- *Defiers*: eat pizza *iff* they don't get the card. We assume away.

Punchline: the ITT over everyone will be an average over these and so will be a *watered-down* version of the pizza effect. Specifically,

Backing it out: the algebra

Averaging the ITT over the four types:

$$\begin{aligned}
\text{ITT} &= \underbrace{(\text{ITT} \mid \text{never-taker})}_{=0} \cdot \Pr(\text{never-taker}) \\
&\quad + \underbrace{(\text{ITT} \mid \text{always-taker})}_{=0} \cdot \Pr(\text{always-taker}) \\
&\quad + (\text{ITT} \mid \text{complier}) \cdot \Pr(\text{complier}) \\
&\quad + \underbrace{(\text{ITT} \mid \text{defier})}_{=0,\ \text{by assumption}} \cdot \Pr(\text{defier}) \\
\Rightarrow \quad \text{ITT} \mid \text{complier} &= \frac{\text{ITT}}{\Pr(\text{complier})}
\end{aligned}$$

So we can recover an ITT-for-compliers.

And for compliers, the ITT is the ATE.

From ITT-on-compliers to LATE

One more thing you need to know: Pr(complier) is the "first stage" effect:

$\Pr(\text{complier}) = \mathbb{E}[D_1 - D_0] = \mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]$ (by ignorability of Z), which equals the proportion of compliers (if no defiers).

Putting it together: We call the ATE among the compliers the "local average treatment effect" (LATE):

$$\text{LATE} = \frac{\text{ITT}}{\Pr(\text{complier})} = \frac{\mathbb{E}[Y \mid Z=1] - \mathbb{E}[Y \mid Z=0]}{\mathbb{E}[D \mid Z=1] - \mathbb{E}[D \mid Z=0]}$$

Formalizing this potential-outcomes machinery

Let $Z_i \in \{0, 1\}$ be the encouragement (instrument) and $D_i \in \{0, 1\}$ the treatment.

Potential treatments. D_{1i}, D_{0i}: would you take the treatment if encouraged ($Z = 1$) or not ($Z = 0$). We see $D_i = Z_i D_{1i} + (1 - Z_i) D_{0i}$.

Four principal strata, defined by (D_{0i}, D_{1i}):

	$D_{0i} = 0$	$D_{0i} = 1$
$D_{1i} = 0$	never-taker	defier
$D_{1i} = 1$	complier	always-taker

Monotonicity $D_{1i} \geq D_{0i}$ for all i rules out the red cell.

The assumptions

1. Exclusion: $Y(d, z) = Y_d$: Z affects Y only through D.

- Not implied by Z being random: Z can be random but *still* affect Y through some channel other than D.
- Not testable though sometimes falsifiable.

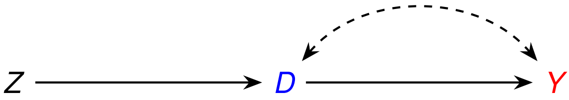
2. Ignorability of Z: $\{Y(d), D(z)\} \perp\!\!\!\perp Z$ for all d, z.

- Ideally Z is literally randomized, or pseudo-randomized (some "shock")

3. First stage / relevance: $\Pr(D_1 = 1) \neq \Pr(D_0 = 1)$ — i.e. there are some compliers. Testable: regress D on Z.

4. Monotonicity: $D_{1i} \geq D_{0i}$ — no defiers. Not testable, but often qualitatively defensible (e.g. a lottery winner is no *less* likely to enroll).

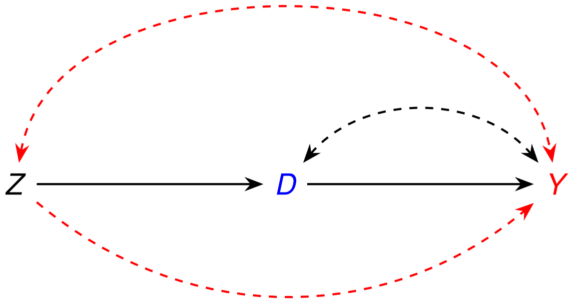
The IV DAG



This would be good.

- Unobserved confounders of D and Y – no problem!
- But what's important is what's not on here...

The IV DAG: edges that should *not* be there



Red edges are the ones an IV story *must* rule out:

- Top: $Z \leftrightarrow Y$: *any* unobserved factor associated with both Z and Y. Violates ignorability of Z.
- Bottom: $Z \rightarrow Y$: a direct path from Z to Y not via D. Violates exclusion.

Estimation: Wald and 2SLS

Wald estimator (sample analog of the ratio):

$$\hat{\beta}_{\text{Wald}} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}$$

Two-stage least squares (2SLS):

1. First stage: $D_i = \pi_0 + \pi Z_i + \eta_i$. Compute $\hat{D}_i$.
2. Second stage: $Y_i = \beta_0 + \beta\,\hat{D}_i + \varepsilon_i$.

With one binary Z and one binary D, $\hat{\beta}_{\text{2SLS}} = \hat{\beta}_{\text{Wald}}$.

In practice: for $\mathbf{X} = [1\ D]$, $\hat{\beta}_{IV} = (Z^\top \mathbf{X})^{-1} Z^\top Y$.

- **In R:** `ivreg(Y ~ D | Z)` from the `AER` package.

Estimation: covariates and weak instruments

Adding covariates when ignorability/exclusion only hold conditional on X:

- Add X to *both* stages.
- With heterogeneous effects, this no longer cleanly identifies a LATE; the complier set varies with X.

Weak instrument diagnostics:

- First-stage F-statistic; rule of thumb $F > 10$.
- Better: Anderson–Rubin confidence intervals — valid even when the instrument is weak.

The ideal use-case: RCTs with non-compliance

Suppose you run a randomized trial,

- Assigned treatment (Z), but D means actual take-up.
- Could be one-way or two-way non-compliance.

You might estimate the ITT, i.e. the effect of *being assigned* to treatment.

- Easy, unbiased, but understates the effect of the treatment itself.

The IV "correction": divide that ITT by compliance rate to get back the "undiluted" effect of actually *taking* the treatment, but only for *compliers*.

- Ignorability of Z: holds *by design* (the experimenter randomized).
- Exclusion: could still be violated but could be defensible – some Ys are hard to move without special treatment.
- Monotonicity: usually defensible.

This is standard fare in many RCTs and field experiments.

Three famous IV Examples

Setup	Instrument Z	Treatment D	Outcome Y
Acemoglu, Johnson & Robinson (2001) *settler mortality*	European settler mortality, c. 1500–1900	quality of institutions today	log GDP per capita
Angrist & Krueger (1991) *quarter of birth*	quarter of birth (Q1 vs. Q3/Q4)	years of schooling	log earnings
Miguel, Satyanath & Sergenti (2004) *rainfall and conflict*	year-on-year rainfall growth	income / GDP growth	civil conflict onset

Who are the compliers in each?
Let's worry about ignorability in each.
Let's worry about exclusion in each.

Application: Vietnam draft lottery (Angrist 1990)

Question: What is the causal effect of military service on later civilian earnings?

The naive comparison (veterans vs. non-veterans) is hopeless: who served depends on health, education, wealth, region, family, motivation ... which may predict earnings.

Angrist's idea: use the Vietnam-era draft lottery (1969–72). Each birthday was randomly assigned a number 1–365; the Selective Service then announced a yearly cutoff, and men with numbers *below* the cutoff were called for induction.

- $Z = 1$ *draft-eligible*: number below the cutoff $\Rightarrow$ called for induction
- $D = 1$: actually served in the military
- Y: civilian earnings, 1981–84

Critical: $Z \neq D$. Among the called ($Z = 1$), only $\sim$35% served (deferments, failed physicals). Among the not-called ($Z = 0$), $\sim$19% still served (volunteers). First stage ≈ 0.16.

Wald estimates from Angrist (1990): white men, 1950 cohort

ITT	($\bar{Y}^e - \bar{Y}^n$, 1981 FICA earnings)	$-\$435.8$	(s.e. 210.5)
First stage	($\hat{p}^e - \hat{p}^n$, veteran rate)	0.159	(s.e. 0.040)
Wald $=$ ITT / first stage	$\frac{-435.8}{0.159} \approx$	$-\$2{,}741$	(s.e. $\approx \$1{,}335$)

Source: Angrist (1990) Table 3, col. 1.

What's the interpretation of this result?

What about the assumptions here?

Ignorability of Z? Yes by design — it's a uniform random number.

First stage / relevance? Lottery eligibility raised the probability of service by $\sim$16 percentage points; $F = (0.159/0.040)^2 \approx 16$. Considered okay.

Monotonicity? Hard to imagine someone served *only* because they got a high (safe) lottery number. Defensible.

Exclusion restriction? e.g. College deferments.

- Men with low lottery numbers were draft-eligible, had incentive to enroll in college to claim a student deferment.
- Thus draft-eligible men ended up with *more* schooling
- ...Z can lead to higher earnings via schooling, not only via service.

LATE for whom? Lottery-induced compliers: those who served *because* of their number. Not always-takers (volunteers), not never-takers (medically deferred).

Rules of the game

1. Show Relevance: Z moves D. *Look at the first stage.*
2. Ignorability + exclusion: ignorability is given by random Z; still worry about exclusion. *Not verifiable but explain why you believe it. You can show certain (observed) channels can be ruled out.*
3. Monotonicity: no defiers. *Often you can argue this on qualitative grounds.*
4. Interpretation: LATE for compliers. *Who are the compliers? Are they policy-relevant?*